Individual Contribution Report

Design and Creative Technologies

Torrens University, Australia

Student: Luis Guilherme de Barros Andrade Faria – A00187785

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Assessment No.: 3

Title of Assessment: Case Study Report

Lecturer: Dr. Nandini Sidnal

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1. Group Details

Student ID	Name	Email
A00187785	Luis Guilherme de Barros Andrade Faria	Luis.Faria@Student.Torrens.edu.au
A00106473	Samiran Shrestha	Samiran.Shrestha@Student.Torrens.edu.au
A00179705	Victor Javier Dorantes Meneses	Victor.Meneses@Student.Torrens.edu.au

- Source code: https://github.com/lfariabr/review-pulse
- Presentation: ISY503_Faria_L_Assessment_3.pptx
- Live Demo: https://review-pulse.streamlit.app/

2. Team Contribution Table

Member	Main contribution	%
Luis	Technical architecture, implementation, integration, testing, documentation, release management, and coordination	50%
Samiran	Problem framing, live demonstration support, and future improvement planning	25%
Victor	DistilBERT/model experimentation and integration support, dataset analysis, error analysis, ethics presentation	25%

3. Individual Report

My primary contribution to ReviewPulse was the full technical implementation and hardening of the ML pipeline - the modular Python package structure, the Streamlit web app, the automated test suite, and the project documentation.

On the data side, I built the pseudo-XML parser (*src/data/parser.py*), the preprocessing pipeline with negation expansion (*src/data/preprocess.py*), and the vocabulary and PyTorch DataLoader helpers (*src/tokenization/*). I then implemented the TF-IDF + Logistic Regression baseline (*src/training/baseline.py*) and the bidirectional LSTM (Hochreiter & Schmidhuber, 1997) initialised with GloVe 100-dimensional embeddings (*src/models/bilstm.py*). The BiLSTM training loop (*src/training/bilstm.py*) uses Adam with gradient clipping, F1-based checkpointing, and Apple MPS device support. I also hardened the final modular architecture: app services in *src/app/service.py*, inference split across *src/inference/*, and evaluation split across *src/evaluation/*. I produced the presentation outline, 10 acceptance test cases with real model outputs, the submission checklist, and this document.

An important ethical consideration is that the Blitzer et al. (2007) dataset uses filename-derived labels rather than human raters. We audited for possible rating/text conflicts and ambiguous boundary cases and found zero ambiguous rows, but the risk remains relevant in broader sentiment classification settings. BiLSTM and DistilBERT confidence values are uncalibrated (high confidence does not imply high accuracy) and both models generalise poorly to out-of-distribution text such as logistics reviews (Bender et al., 2021). Any production deployment requires human oversight and periodic label audits.

I estimate my contribution at 50% as the primary technical implementer and final refactor lead. Victor contributed 25% covering DistilBERT/model experimentation, integration support, dataset analysis, error analysis, and ethics. Samiran contributed 25% through supported problem framing, live demonstration support, and future improvement planning, including review-classification challenges such as negation, sarcasm, review length, and domain differences.

4. Appendices

4.1 Appendix A – Github Repository

https://github.com/lfariabr/review-pulse

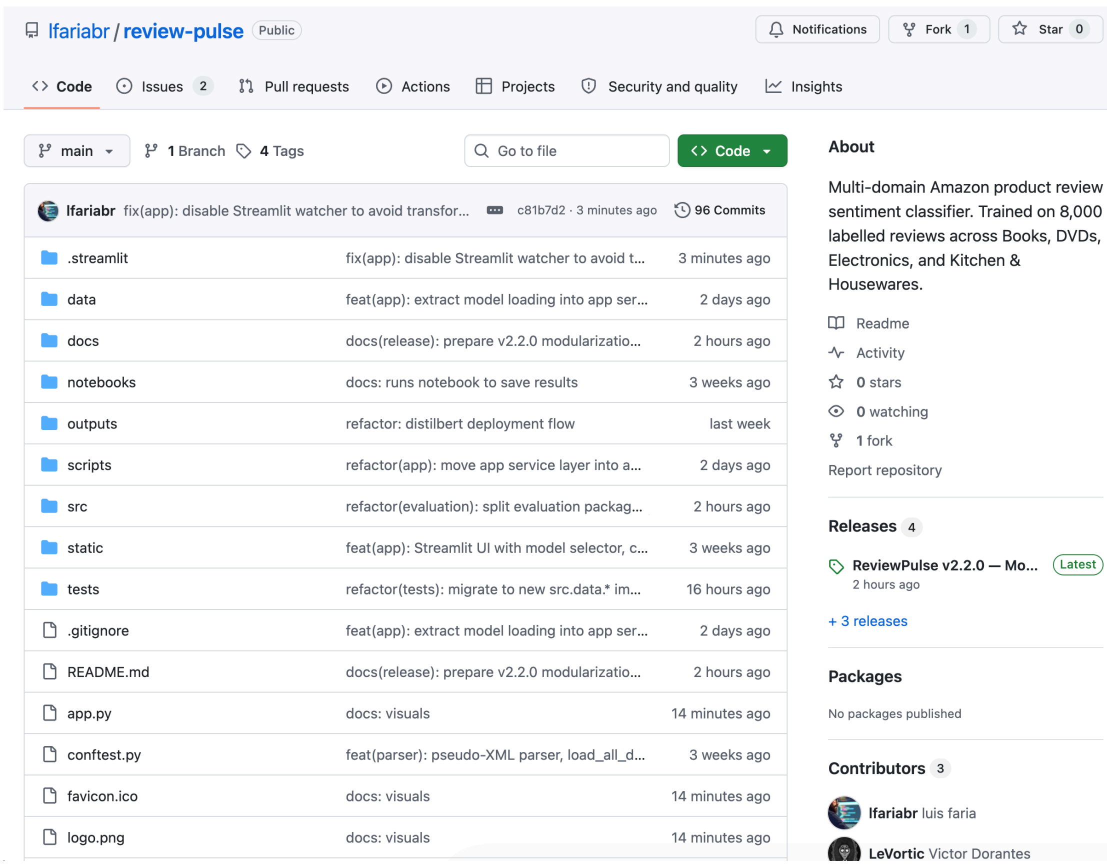

Fig A1. Github repository main page.

4.2 Appendix B – Live Application

https://review-pulse.streamlit.app/

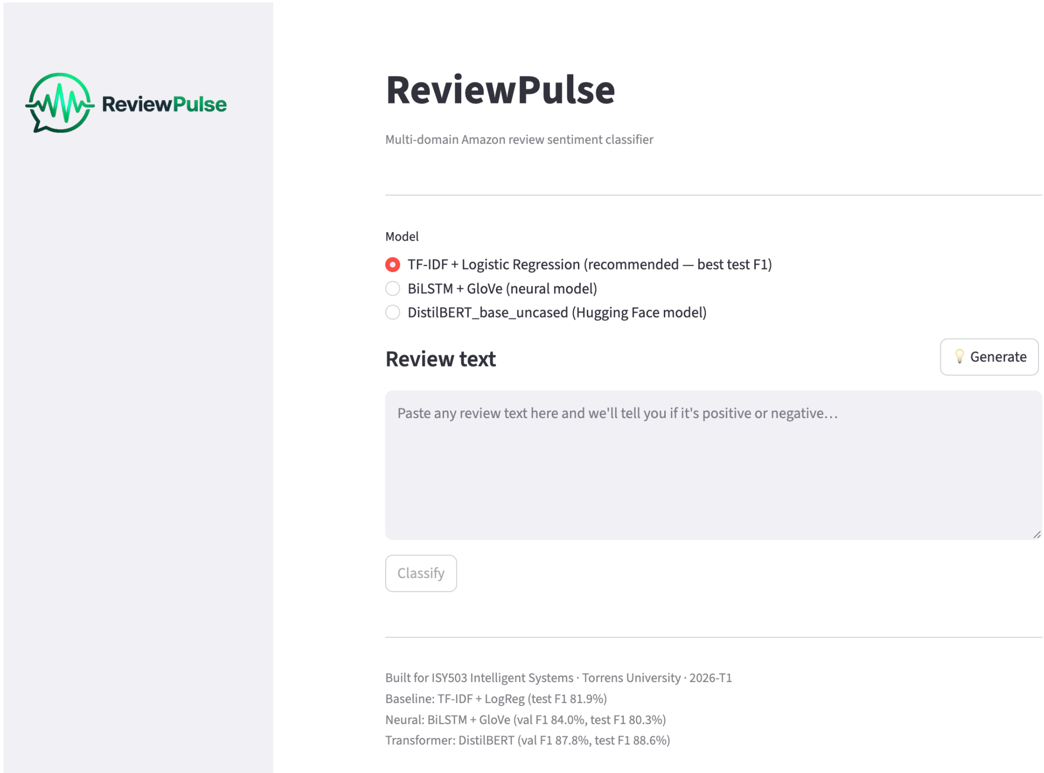

Figure A2.1. ReviewPulse app homepage with model selector.

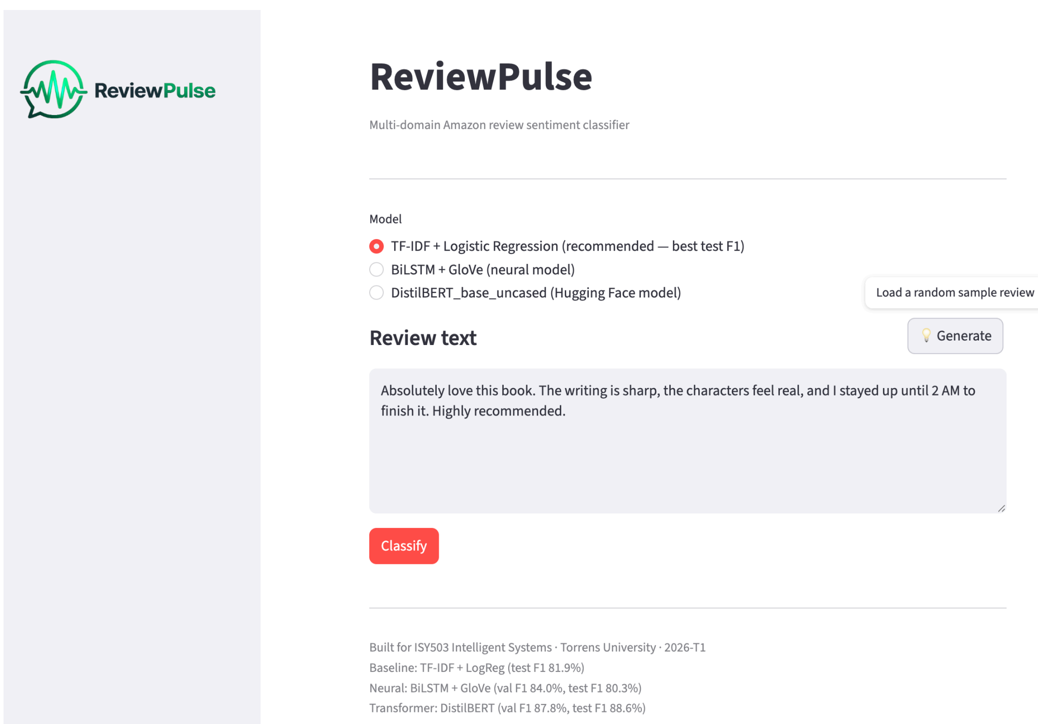

Figure A2.2. ReviewPulse app with random sample review generated and ready to classify.

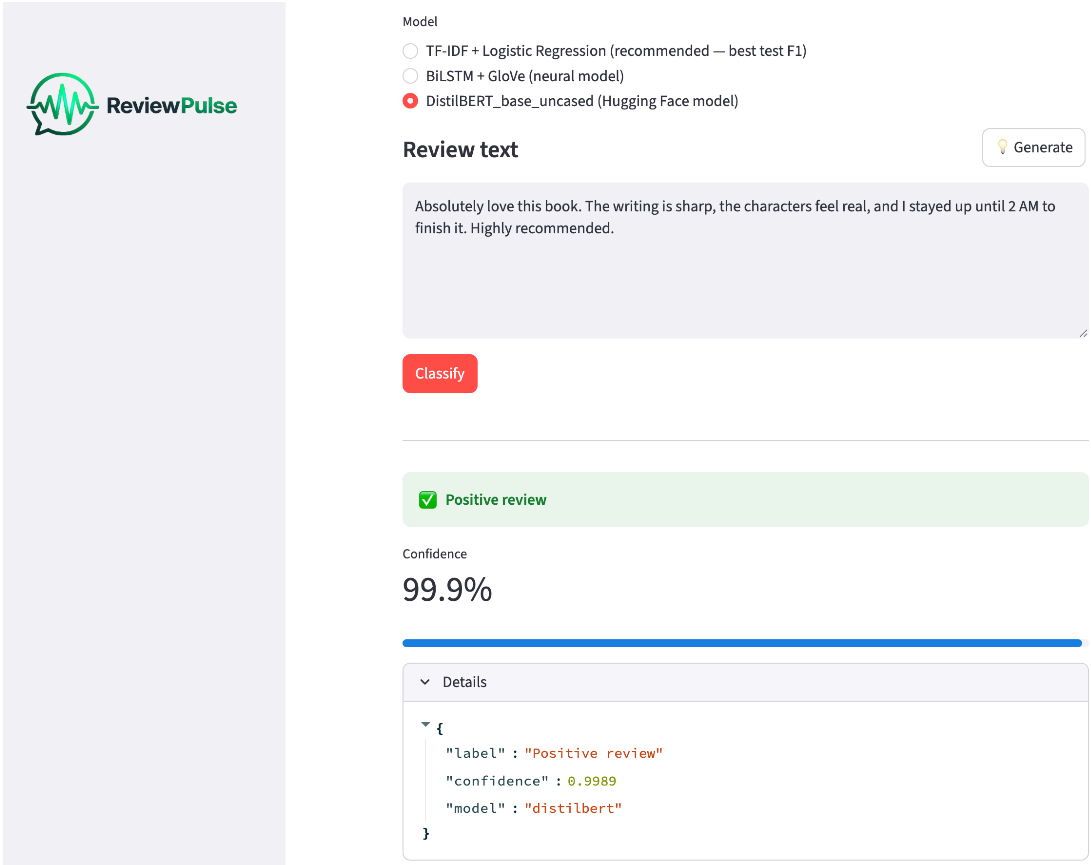

Figure A2.3. ReviewPulse app with input review text classified.

3.3 Appendix C – Sample Test Cases Output

Ten acceptance test cases run against the baseline and BiLSTM checkpoints (baseline checkpoint *outputs/baseline.joblib*, BiLSTM checkpoint *outputs/bilstm.pt* epoch 9). DistilBERT is integrated into the app as the third selectable model using *outputs/distilbert.pt*.

Case	Input (excerpt)	Baseline	BiLSTM
Clear positive	This blender is absolutely incredible…	Positive 73.8%	Positive 97.9%
Clear negative	Broke after two days…	Negative 95.9%	Negative 99.6%
Negation trap	This is not bad at all…	Negative 66.9%	Negative 74.6%
Sarcasm	Oh great, stopped working after a week…	Negative 52.5%	Negative 64.6%
Domain-shifted (books)	One of the best thrillers I have read…	Positive 69.4%	Positive 86.2%

Table A3. Sample Test Cases Output.

Full results: https://github.com/lfariabr/review-pulse/blob/main/docs/assessment-files/demo-test-cases.md.

3.4 Appendix D – Future Work

Priority	Extension	Rationale
Shipped	DistilBERT	Victor implemented Hugging Face *distilbert-base-uncased*; test F1 88.6%, live in app
High	RoBERTa	Larger pretrained transformer; expect to push F1 beyond 90%
High	Confidence calibration (Platt scaling)	BiLSTM logits are uncalibrated; 98% confidence ≠ 98% accuracy
Medium	Additional training domains	Current model is trained on 4 product categories only; broader domains would validate generalisation
Medium	LIME / attention visualisalion	Makes predictions auditable - necessary for any production or high-stakes deployment
Low	FastAPI backend	Decouple inference from Streamlit for easier integration with external systems

Table A4. Future Work possibilities and opportunities.

End of Appendix Section

Statement of Acknowledgment

I acknowledge that I have used the following AI tool(s) in the creation of this report:

- OpenAI ChatGPT (GPT–5.4)
- Anthropic Claude Opus 4.6

Both tools were used to assist with understanding ML concepts, structuring the technical manual, improving clarity of academic language, and supporting APA 7th referencing conventions.

Prompt examples:

1. *"I built a BiLSTM sentiment classifier with pack_padded_sequence to stop the LSTM processing padding tokens. My training achieves val F1=84.0% at epoch 9, but test F1 drops to 80.3% while the TF-IDF baseline reaches 81.9% on the same held-out set. Can you help me understand why the neural model didn't generalise better despite the stronger validation score?"*
2. *"I need to compare two approaches for sentiment analysis – TF-IDF + Logistic Regression vs BiLSTM with GloVe – in an academic report where the simpler model wins on the held-out test set. How do I frame this honestly without undermining the neural architecture requirement of the assessment rubric?"*
3. *"Format this as APA 7th: Bender, Emily M., Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell, 2021, article titled On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?, conference ACM FAccT 2021, pages 610 to 623."*

I confirm that the use of these tools has been in accordance with the Torrens University Australia Academic Integrity Policy and TUA, Think and MDS's Position Paper on the Use of AI. I confirm that the final output is authored by me and represents my own critical thinking, analysis, and synthesis of sources. I take full responsibility for the final content of this report.

5. References

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM FAccT Conference* (pp. 610–623). https://doi.org/10.1145/3442188.3445922

Blitzer, J., Dredze, M., & Pereira, F. (2007). Biographies, Bollywood, Boom-boxes and Blenders: Domain adaptation for sentiment classification. In *Proceedings of the 45th Annual Meeting of the ACL* (pp. 440–447). ACL. https://aclanthology.org/P07-1056/

Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation, 9* (8), 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735